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Applied Analytics Project - Week 4 Assignment

Dataset Cleaning/Processing:

Prior to this week's assignment, the majority of data cleaning and preprocessing for our March Madness dataset had already been completed during earlier exploratory analysis. As a result, this week primarily focused on validating prior decisions and confirming additional transformations were unnecessary for effective modeling.

Initial cleaning steps focused on removing variables that contained a large proportion of missing values. Specifically, several columns related to positional height averages and certain offensive and defensive ratings were removed because of a number of missing values across the teams and seasons. We also removed columns that were not appropriate for predictive modeling: coach name, conference, and binary indicator variables such as "Top 12 in AP Top 25 Week 6". After these removals, the remaining columns were confirmed to contain no missing values. Therefore, no additional missing data treatment or imputation was necessary.

A major transformation performed last week was converting all team statistics into difference based features. Instead of modeling raw statistics for both teams, we took the difference of each statistic for Team 1 and Team 2. This transformation is appropriate for head to head matchups because the outcome is dependent on relative team strength rather than absolute variables. Using differences reduces redundancy and helps prevent multicollinearity.

Outlier analysis indicated that extreme values in team statistics were rare and typically reflect legitimate performance differences between teams. Because these values represent true advantages rather than data entry errors, we chose not to manipulate them. In a tournament setting, extreme performance differences may be highly predictive and should be preserved as much as possible.

We elected not to normalize or standardize the predictor variables for several reasons. The magnitude of difference between two teams is inherently meaningful. Larger differences in many team statistics may directly indicate a higher probability of winning. Scaling these values could reduce interpretability and potentially decrease meaningful variance. Also, most team metrics are approximately normally distributed due to the large sample of Division I teams and the nature of competitive play. When converted to difference based features, the distributions remain centered which reduces the need for scaling. Finally, since larger absolute differences may carry stronger predictive power, scaling could unintentionally compress important distinctions between matchups. For these reasons, we determined that normalization or standardization would not meaningfully improve performance and could potentially obscure valuable information.

In summary, the dataset required minimal additional processing during this week because of the cleaning and feature engineering that had already been performed previous weeks. We removed variables with many missing values, eliminated non-informative categorical features, engineered difference based predictors to create specific matchups, and determined that normalization was unnecessary. The final dataset is clean, structured, and directly optimized for binary classification modeling of March Madness matchup outcomes.